John C. Glenn II

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EDUCATION

M.S. Mechanical Engineering | NC State University | GPA: 4.0 — May 2026

B.S. Mechanical Engineering | NC State University | GPA: 3.65 — May 2025

Undergraduate Coursework | UNC Wilmington | GPA: 3.97 — August 2020 - May 2022

EXPERIENCE

NC State Plant Sciences Initiative | Mechanical Engineering Intern — Raleigh, NC | May 2025 - Present

- Led the mechanical design and control of a self-driving farm robot, with a 3-dof gantry and modular end effector
- Built and deployed tractor-pulled imaging system using low-cost edge computing and cameras to collect data for training the staking machine's computer vision system

Gamewell Mechanical | Mechanical Engineering Intern — Salisbury, NC | May 2024 - August 2024

- Identified an inefficiency in ductwork labeling and designed and implemented a custom tool that reduced process time and labor costs by over 50% while eliminating the need for scissor lifts.

Martin Marietta Materials | Mechanical Engineering Intern — Castle Hayne, NC | May 2023 - August 2023

- Designed a diverting chute to support plant expansion using SolidWorks which improved efficiency and reduced downtime by redirecting flow of material during maintenance periods

PROJECTS

Autonomous Tomato Harvester

- Developed a multi-input multi-output (MIMO) LQG control system using optimal control theory for a 4-dof mechanical arm
- Achieved end effector positioning accuracy of ±2mm by utilizing a gain scheduled LQR controller for optimal performance at various operating points and a Kalman filter for optimal state estimation
- Applied grey-box system identification techniques to model system dynamics from experimental data
- Identified ripe tomatoes and position data using computer vision, convolutional neural networks, and triangulation
- Applied design for manufacturing (DFM) principles during CAD design and fabrication of components on mill

Double Inverted Pendulum on a Cart: Swing-up and Stabilization

- Developed both passivity-based partial feedback linearization and offline trajectory optimization (trapezoidal direct collocation) for underactuated swing-up control, reducing swing-up time from 19.1s to 4.3s
- Designed and tuned online model predictive controller (MPC) to track trajectory and stabilize the pendulum once within the domain of attraction (DoA) at desired equilibrium points

Spacecraft Orbit Determination and State Estimation

- Developed an Iterated Square Root Information Filter (SRIF) and Extended Kalman Filter (EKF) from scratch in MATLAB to perform orbit determination for a spacecraft with unknown orbital parameters and dynamics
- Estimated spacecraft states, impulsive maneuvers, J2/J3, drag, lunar third-body effects, and station positions given only simulated range and range-rate measurements and *a priori* estimate

Inverted Pendulum for Controls Curriculum

- Engineered and fabricated an inverted pendulum on a cart for NCSU's controls curriculum, prioritizing robustness to withstand repeated student use and controller failures
- Designed CAD model and shop drawings in SolidWorks utilizing GD&T standards and DFM principles
- Implemented sliding mode control using Simulink Desktop Real-Time (SDRT) and modeled system dynamics

Robotic Manipulator Simulation and Control

- Simulated pick-and-place manipulator in Mujoco/Python, modeling contact dynamics for reliable grasping
- Implemented inverse kinematics to map desired end-effector poses to joint space configurations

SKILLS

GNC: LQR, SMC, MPC, state estimation (EKF, SRIF), trajectory optimization, system identification, motor control, inverse kinematic (IK) control, spacecraft navigation

Software: MATLAB, Simulink, SDRT, Python, Mujoco, SolidWorks, Fusion360

Hardware: oscilloscope, signal generator, multimeter, soldering, microcontrollers, vertical mill, welding, 3D printing